

Response to Reviewers

Paper: How Much Does the Activation Function Matter? A 2,000-Run Variance Study of Gated Recurrent Networks for Sentiment Classification

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We thank both reviewers. Reviewer 2 and Reviewer 3 independently identified the same weakness, that the sweep was single-seed. Following that point changed the paper substantially.

We did not restrict the replication to the top ten configurations as asked. We re-ran **every one of the 400 configurations under five seeds, for 2,000 trained models**, because a partial replication could not have told us whether the problem was local to the leaders. It was not. The replication showed that the single-seed ranking of activation pairs does not reproduce in any of the four architectures, which we now report as the paper’s principal finding. Several claims in the submitted version did not survive and have been removed or reversed; these are listed explicitly at the end of this letter.

The title, abstract, research questions, results and conclusion have been revised accordingly. The experimental design, architectures, hyperparameters and reference list are unchanged.

Reviewer 2

Methodology sound. Primary reason for revision: address single-seed statistical variance in the results section.

Addressed. See Section III-E (Replication Protocol), Section IV-A, Table IV, and Table VI.

Every configuration was retrained under five seeds {10, 20, 30, 40, 50} with the data split held fixed at `random_state=42`, so the reported variance is the variance of retraining the same model on the same data. That is the quantity a single-run comparison implicitly assumes to be negligible.

A variance decomposition over the 2,000 runs (Table IV) gives:

Source	% of total variance
Random seed	58.67
Architecture	26.69
of which cell type (GRU vs. LSTM)	26.32
of which layer-2 bidirectionality	0.19
Act ₁ × Act ₂ pair	3.74

The seed accounts for more variance than every controlled factor combined, and roughly sixteen times more than the activation pair. Two conclusions changed as a result:

- **The top-ten ranking is withdrawn.** The ten best of 400 configurations span 0.12 accuracy points against confidence half-intervals up to ± 0.50 . Every pair of intervals overlaps. Table V now reports them as statistically indistinguishable and the text states that the ordering must not be read as a ranking.
- **The architecture conclusion is strengthened.** GRU exceeds LSTM by 0.79 points over 1,000 runs per family ($p \approx 2 \times 10^{-125}$, Cohen’s $d = 1.19$), up from the 0.73 points reported from single runs. We additionally report that GRU is the more *reliable* family, with roughly 40% lower seed-to-seed standard deviation ($p \approx 2 \times 10^{-7}$), and that all eight high-variance cells in the grid are LSTM-based.

We also added a power analysis (Section V-F) giving the seed budget needed to resolve each effect: about 4 seeds per cell for the architecture difference, about 157 to separate the best cell from the tenth. This makes the limitation quantitative rather than rhetorical.

Reviewer 3

Point 1: Figure quality

Regenerate Fig. 1 and Fig. 2 as vector (PDF/EPS) or ≥ 300 DPI raster. All axis titles, ticks, bar labels fully legible, no clipping.

Addressed. Both figures regenerated as vector PDF.

Both are now produced by a script that embeds TrueType subsets (`pdf.fonttype = 42`), reads the trimmed bounding box back out of the written PDF, and verifies the on-page size of the smallest text. Fig. 1 renders at 8.6 pt and Fig. 2 at 8.3 pt, above the 8 pt floor, with no clipped labels and legends placed clear of the data.

Fig. 2 (formerly Fig. 1) is the convergence plot; the bar chart that was Fig. 2 has been **replaced rather than redrawn**. The submitted version plotted the single best activation pair per architecture; those peaks are maxima of a noisy single-seed sweep and do not reproduce, so redrawing them would have made a misleading figure merely legible. The new Fig. 1 plots all 400 replicated cell means grouped by architecture with 95% confidence intervals, which shows directly what the data supports: the architectures separate, the activation pairs within them do not.

Fig. 2 now shows the mean validation-loss trajectory over five seeds with ± 1 standard deviation bands, rather than a single run.

Point 2: Error bars and confidence intervals

Main sweep is single-seed. Add standard deviation error bars or confidence intervals for the top-10 configurations across multiple seeds to show differences are statistically meaningful.

Addressed, and extended to all 400 configurations. See Table V, Table VI and Fig. 1.

Table V reports the ten best configurations with mean, standard deviation and 95% confidence interval, as requested. The direct answer to the reviewer’s question is that the differences are **not** statistically meaningful: the ten means span 0.12 points and all intervals overlap.

Because we replicated the whole grid rather than only the leaders, we can also report how far the single-seed ranking is from the replicated one (Table VI):

Experiment	Spearman ρ	Mean rank Δ	Top-10 overlap
BiLSTM+LSTM	−0.174	36.9	0/10
BiGRU+GRU	−0.059	34.5	1/10
BiLSTM+BiLSTM	−0.041	34.4	1/10
BiGRU+BiGRU	+0.055	32.1	0/10

Four independently swept architectures, four correlations indistinguishable from zero. We report this as the paper’s main result.

We chose tables over error bars on the figure for the numeric comparisons because Table V must carry four quantities per row (mean, σ , interval, F1) for ten rows; error bars would convey the overlap qualitatively but not the interval widths, and the tabular form costs less space under the six-page limit. Fig. 1 does carry confidence intervals, for the architecture means, where a graphical comparison is the more useful presentation.

Point 3: Additional dataset

(Consider) Test top 5–10 pairings on an additional sentiment dataset (SST-2 / Amazon) to confirm the LSTM/GRU inversion holds across domains.

Not carried out. Named as the specific next step in the Conclusion.

We did not have the budget to add a second corpus within the camera-ready window, having spent it on the 2,000-run replication. We are not promising it here; we state plainly that it is untested.

One clarification is owed to the reviewer, because the premise of the suggestion no longer holds. The cross-architecture *inversion* of activation rankings that this point asks us to confirm is not a real effect. It was an artefact of single-seed measurement. On replicated data the six pairwise correlations between architectures’ activation

marginals range from -0.164 to $+0.576$ and none is significant ($p \geq 0.082$): the rankings neither transfer nor invert. There is consequently nothing to confirm across domains on that specific point, and the claim has been removed from the paper.

What a second corpus would genuinely test is whether the variance split and the architectural advantage generalise, and the Conclusion now names SST-2 and Amazon Reviews for exactly that purpose. We agree it is the right next step.

Additional corrections made without prompting

1. **Fig. 1 caption was wrong.** It read “Per-Epoch Validation Accuracy” while the plot showed validation *loss*. Corrected.
2. **A contradiction over seeds is resolved.** The submitted Conclusion attributed two results (88.48%, 88.43%) to “a different seed in the convergence rerun”, while Table VII stated the seed was fixed at 42. Auditing the code showed the opposite of what the text implied: the convergence rerun *was* seeded at 42, and the main 400-run sweep set *no seed at all*. Both statements have been removed. Over 2,000 seeded runs no configuration’s mean reaches the 88.40% benchmark and no individual run attained it (maximum 88.39%), so the claim that the benchmark “sits within normal seed variance” is not supported and has been withdrawn.
3. **Table IV row labels were corrupted** in the submitted source (“BiLSTM+BiLSTM+LSTM”, “BiGRU+BiGRU+GRU”). Corrected.
4. **Claims withdrawn as unsupported by the replicated data.** Listed here for transparency: the identity of the single best activation pair; the cross-architecture inversion of rankings; the per-configuration precision–recall imbalances of the former Table VIII (the widest gap between mean precision and mean recall over any cell is 6.89 points, while individual runs reach 24.17, so these are properties of runs, not of configurations); and the dead-neuron penalty for dense-layer ReLU of the former Table X (no run of 2,000 fell below 60% accuracy, and in BiLSTM+LSTM ReLU is in fact the strongest Act₁ choice, $p = 0.006$). The corresponding tables have been removed and the space used for the variance and reproducibility results.

The paper is within the six-page limit, uses the IEEE conference template, carries the required first-page header and copyright footer, and has all fonts embedded. The single-seed observation was the most useful comment we received; acting on it in full produced a stronger result than the one we submitted.